DO NOW — RECALL

Lesson 124 · Procedural Programming: Modules and Sta

Name: _____ Class: _____ Date: _____

1. Which paradigm describes a desired result, rather than detailed step-by-step changes?

2. A full adder receives A=1, B=1 and carry-in=0. What are sum and carry-out?

3. For the binary value 1010, what is its decimal value?

4. A linked-list node has data and what other essential field?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Declarative programming.

2. Model answer

Sum 0 and carry-out 1.

3. Model answer

10.

4. Model answer

A link/reference to the next node (for a singly linked list).

By the end, you can…

• Statements execute in an order; selection and loops direct flow.

• Reusable named routines accept parameters; functions return values.

• Local variables limit unwanted side effects; global state needs care.

Vocabulary — know these terms

procedure — Named subroutine mainly performing an action.

function — Subroutine that returns a value.

parameter — Named input used by a routine.

scope — Region where an identifier may be accessed.

NEW LEARNING

Read, annotate and explain

Sequence and control

Statements execute in an order; selection and loops direct flow.

Procedure/function

Reusable named routines accept parameters; functions return values.

Scope and state

Local variables limit unwanted side effects; global state needs care.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace FUNCTION doublePlusOne(x): result=x*2; RETURN result+1; print(doublePlusOne(4)).

Model answer

Parameter x=4, local result=8, return value 9, printed output 9. The local result does not become a global variable merely because the function runs.

Check: link each claim to the scenario and use precise OCR terminology.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain the benefits of using functions/procedures rather than one long main routine. [6]

Question 2

Trace FUNCTION f(n): a=n+2; RETURN a*3; OUTPUT f(2). [4]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain local/global scope and give a bug caused by modifying shared global state. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A local variable declared inside a function is automatically accessible everywhere in the program.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.